

TASVIRLARNI TAHLIL QILISH VA ANGLASH · Amaliy mashg'ulot (Practice)

17-mavzu. Chastota fazosi: Furye almashtirishi

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Muhit. Python 3.11/3.12, faqat CPU. Versiyalar qat'iy belgilangan (to'liq ro'yxat — requirements.txt): numpy==2.4.4, scipy==1.17.1, opencv-python-headless==4.13.0.92, scikit-image==0.26.0. Tasvirlar skimage.data ichida (camera, astronaut) yoki sintetik — internet kerak emas.

```
python -m venv .venv && source .venv/bin/activate # Windows: .venv\Scripts\activate
pip install -r requirements.txt
# har bir kodli topshiriq boshida:
import numpy as np, cv2
from scipy import signal
from scipy.fft import dctn, idctn
from skimage import data, color
from skimage.metrics import structural_similarity as ssim
SEED = 42; rng = np.random.default_rng(SEED)
img = data.camera().astype(np.float64) # 512x512
psnr = lambda a, ref: 10 * np.log10(255**2 / np.mean((a - ref) ** 2))
u, v = np.mgrid[:512, :512]; D = np.hypot(u - 256, v - 256) # fftshift qilingan spektrda masofa
```

Mavzu. 1D/2D DFT va IDFT, FFT (Cooley–Tukey), DFT xossalari (siljish, konjugat simmetriya, Parseval, ajraluvchanlik), convolution theorem, zero-padding, low-pass/high-pass (ideal, Butterworth, Gaussian), ringing, high-boost, notch filter, inverse va Wiener filtri, phase correlation, DCT, aliasing, noaniqlik munosabati.

Maqsadlar. Talaba: (1) kichik signal va tasvirlar uchun DFT, konvolyutsiya va filtr qiymatlarini qo'lda hisoblaydi; (2) DFT xossalari, convolution theorem va FFT rekursiyasini isbotlaydi; (3) NumPy/SciPy/OpenCV bilan chastotaviy filtrlarni quradi va natijani PSNR/SSIM bilan baholaydi; (4) davriy shovqin, xiralashish va siljish masalalarini chastota fazosida yechadi.

Baholash (100 ball). A (1–14): $14 \times 2 = 28$; B (15–18): $4 \times 3 = 12$; C (19–30): $12 \times 4 = 48$; hisobot sifati: 12. Oraliq qadamlarsiz javob — yarim ball. Kod `python -W error::DeprecationWarning` bilan xatosiz ishlashi shart.

A. Qo'lda hisoblash ($X_k = \sum_n x_n e^{-2\pi i k n / N}$; D — markazdan masofa)

- 1D DFT.** $x = (2, 0, 1, 3)$ ning DFT sini toping; konjugat simmetriya va Parseval tengligini tekshiring.
- IDFT.** $X = (8, 2 - 2i, 0, 2 + 2i)$ ($N = 4$) bo'yicha x_n ni tiklang. Nega x haqiqiy chiqadi?
- 2D DFT.** $I = \begin{pmatrix} 4 & 0 \\ 2 & 2 \end{pmatrix}$ uchun $F(k, l)$ ni toping. $F(0, 0)$ ning ma'nosi nima?
- Chiziqli naqsh.** 4×4 tasvir: 0- va 2-qatorlar $(1, 1, 1, 1)$, 1- va 3-qatorlar 0. 2D DFT ni ajraluvchanlik bilan toping. Tasvir bir qatorga pastga doiraviy surilsa F qanday o'zgaradi?
- Siljish xossasi.** $x = (1, 2, 3, 4)$ ning DFT si $(10, -2 + 2i, -2, -2 - 2i)$. Doiraviy siljigan $x' = (4, 1, 2, 3)$ ning DFT sini xossadan foydalanib yozing va $|X'_k| = |X_k|$ ekanini ko'rsating.
- Convolution theorem.** $x = (1, 2, 0, 1)$, $y = (1, -1, 0, 0)$. Doiraviy konvolyutsiyani to'g'ridan-to'g'ri va $X_k Y_k$ ning IDFT si orqali toping. Chiziqli konvolyutsiya nima, u qaysi P da doiraviy bilan ustma-ust tushadi?
- Padding va FFT narxi.** 480×640 tasvir va 31×31 yadro: chiziqli konvolyutsiya uchun minimal $P \times Q$ va 2, 3, 5 ko'paytuvchili eng yaqin katta o'lchamlar. $N = 1024$ da DFT va radix-2 FFT kompleks ko'paytmalari soni.
- Davriy naqsh.** 256×256 tasvir $I = 128 + 50 \sin(2\pi(8u/256 + 16v/256))$. Spektr cho'qqilari qaysi (k, l) da (fftshift dan keyin qaysi piksellarda)? Naqsh davri (piksel) va to'lqin vektori burchagini toping.
- Filtr qiymatlari.** $D_0 = 40$. $D = 10, 40, 100$ uchun Butterworth low-pass ($n = 1$ va $n = 2$), Gaussian low-pass $e^{-D^2/(2D_0^2)}$ va mos high-pass qiymatlarini jadvalga yozing.
- Box filtr javobi.** $N = 8$, $h = \frac{1}{5}(1, 1, 1, 0, 0, 1, 1)$ (markazi $n = 0$). H_k ni hisoblang. Qaysi k da $H_k < 0$? Bu Gauss filtriga nisbatan nimani anglatadi?
- Inverse va Wiener.** $K = 0,02$. $H = 0,2$ va $H = 0,02$ chastotalarda inverse $(1/H)$ va Wiener $(H/(H^2 + K))$ kuchaytirishini toping.
- Aliasing.** $x_n = \cos(2\pi \cdot 0,35 n)$, $N = 200$. DFT cho'qqisi qaysi k da? Har ikkinchi namuna olinsa ($N = 100$), yangi chastota (sikl/namuna) va cho'qqi indeksi qanday? Nyquist chegarasi bilan tushuntiring.
- DCT.** $x = (4, 4, 2, 2)$ uchun ortonormal DCT-II $C_k = \alpha_k \sum_n x_n \cos \frac{\pi(2n+1)k}{2N}$ ni toping. Energiyaning qancha qismi birinchi ikki koeffitsientda?

14. **Noaniqlik.** $N = 256$ da fazoviy Gauss $\sigma = 4$ ning chastotadagi $\sigma_k = N/(2\pi\sigma)$ va FWHM ini toping. U qaysi D_0 li Gaussian low-pass ga teng?

B. Isbot va konseptual savollar

15. **Xossalar.** Siljish xossasi $x_{n-n_0} \leftrightarrow X_k e^{-2\pi i k n_0 / N}$ va haqiqiy x uchun $X_{N-k} = \overline{X_k}$ ni isbotlang.
16. **Convolution theorem.** Doiraviy konvolyutsiya uchun $x \otimes y \leftrightarrow X_k Y_k$ ni isbotlang. Nega $P \geq A + B - 1$ uzunlikkacha nol bilan to'ldirish chiziqli konvolyutsiyani beradi?
17. **FFT.** Radix-2 butterfly $X_k = E_k + W_N^k O_k$, $X_{k+N/2} = E_k - W_N^k O_k$ ni keltirib chiqaring va $T(N) = O(N \log N)$ ni ko'rsating. $x = (1, 2, 3, 4)$ uchun qadam-baqadam bajaring.
18. **Konseptual.** (a) Nega ideal low-pass ringing beradi, Gaussian esa yo'q? (b) Nega tasvirning shakli asosan fazada kodlangan? (c) Nega spektrni ko'rsatishda `np.log1p(np.abs(F))` ishlatiladi?

C. Kodli topshiriqlar (har biri alohida funksiya; ko'rsatilgan formatda chop eting)

19. **DFT va FFT.** `dft_matrix(x)` ($O(N^2)$, $W_{kn} = e^{-2\pi i k n / N}$) va rekursiv `fft_radix2(x)` ni yozing. `x=rng.normal(size=256)` uchun `np.fft.fft` bilan farq $< 10^{-9}$ ekanini (True/False), (1, 2, 3, 4) natijasini va $N \in \{256, 1024, 4096\}$ uchun ko'paytmalar sonini chop eting.
20. **2D xossalar.** `img` uchun: `F(0,0)` va `img.sum()`; Parseval nisbati (12 xona); konjugat simmetriya (nisbiy xato $< 10^{-12}$); qator-ustun 1D FFT = `fft2`; `np.roll(img, (40, -25))` da $|F|$ o'zgarmasligi; log-spektr markazi, maksimumi, o'rtachasi (3 xona).
21. **Convolution theorem.** 15×15 Gauss ($\sigma = 3$, `cv2.getGaussianKernel`) ni markazi (0,0) ga ke-ladigan qilib joylab (`np.roll`) FFT orqali doiraviy konvolyutsiya qiling. `cv2.filter2D` bilan: (a) `np.pad(img, 7, mode='wrap')` dan keyin; (b) sukut chegarasi bilan solishtiring: ichki $[20: -20]$ va chekkadagi farqlar, farqi > 1 piksellar soni.
22. **Zero-padding.** `x=rng.normal(size=100)`, `y=rng.normal(size=21)`: `linear_conv_fft(x,y)` ($P = 120$, `np.fft.rfft`) ni `np.convolve` bilan solishtiring; $P = 100$ da nechta had noto'g'ri va qaysi indekslarda? 2D: `signal.fftconvolve` va `signal.convolve2d` (31×31 , $\sigma = 6$, `mode='full'`); `cv2.getOptimalDFTSize(542, 510, 670)`.
23. **Davriy naqsh.** $N = 256$: `128+50*sin(2*pi*(k0*u/N+10*v/N)) + rng.normal(0,20)` ni $(k_0, l_0) \in \{(8, 16), (-10, 25), (0, 32)\}$ uchun yarating (bitta `rng`). `find_peak(img)` (DC atrofi $r < 3$ va chap yarim tekislik nollanadi) bilan (k, l) , davr va burchakni chop eting.
24. **Low-pass taqqoslash.** $D_0 = 40$: ideal, Butterworth ($n = 2$), Gaussian. Har biri uchun PSNR, SSIM (`data_range=255`) va osmon sohasi $[40:120, 330:500]$ dagi std; Butterworth $n = 1, 2, 5, 10$ uchun osmon std. Ringingni sonlar bilan izohlang.
25. **High-boost.** Gaussian high-pass $1 - e^{-D^2/(2 \cdot 20^2)}$; natijaning o'rtachasi ($|\cdot| < 10^{-9}$) va std; $H = a + b H_{HP}$ ($a = 1$, $b \in \{0.5; 1; 2\}$), natijani $[0, 255]$ ga kesib, o'rtacha Sobel gradient moduli va PSNR ni chop eting.
26. **Notch filter.** `noisy = img + 25 sin(2pi(20u/512+30v/512)) + 20 sin(2pi(-40u/512+10v/512))`. `notch_auto(noisy, n_pairs=2, D0=4, r_protect=10)`: markazni himoyalab 4 cho'qqini topish, ideal notch bilan o'chirish. Cho'qqilar ro'yxati; shovqinli, notch, GaussianBlur ($\sigma = 2$), medianBlur(5) uchun PSNR va SSIM.
27. **Wiener.** Gauss yadro (31×31 , $\sigma = 2$) bilan doiraviy xiralashtiring, `rng.normal(0,2)` qo'shing. Xira tasvir PSNR, $\min |H|$, inverse filtr ($|H| < 10^{-3}$ bo'lsa 10^{-3}) va Wiener $K \in \{10^{-3}, 10^{-2}, 10^{-1}\}$ PSNR ni chop eting.
28. **Phase correlation.** `b = gain*np.roll(img, sh) + 30 + rng.normal(0,5)`; $(sh, gain) \in \{((-17, 29), 1), ((40, -5), 0.6)\}$. `phase_corr(a,b)->(dy,dx,peak)` va `cv2.phaseCorrelate(a,b)` natijalarini chop eting.
29. **Amplituda va faza.** `camera` va `color.rgb2gray(data.astronaut())*255`: amplituda va fazalarni almashtirib IDFT qiling, har natijaning ikki manba bilan korrelyatsiyasini (`np.corrcoef`) chop eting. Faqat faza ($|F| = 1$) tiklanmasining moduli va Sobel gradient moduli korrelyatsiyasini ham toping.
30. **DCT va JPEG g'oyasi.** `camera` ni 8×8 bloklarga bo'ling, `dctn(..., norm='ortho')`; $u + v < K$, $K \in \{1, 3, 4, 6\}$ koeffitsientlarni saqlab energiya ulushi va PSNR ni hisoblang; DFT da shuncha eng past chastotani ($\min(k, 8 - k)^2 + \min(l, 8 - l)^2$ bo'yicha barqaror saralash) saqlab solishtiring; `cv2.dct` bilan tekshiring.

Hujjatlar

- NumPy FFT: <https://numpy.org/doc/stable/reference/routines.fft.html>
- SciPy `fftconvolve`: <https://docs.scipy.org/doc/scipy/reference/generated/scipy.signal.fftconvolve.html>
- SciPy `dctn`: <https://docs.scipy.org/doc/scipy/reference/generated/scipy.fft.dctn.html>
- OpenCV Fourier Transform: https://docs.opencv.org/4.x/de/dbc/tutorial_py_fourier_transform.html
- OpenCV `dft`, `getOptimalDFTSize`: https://docs.opencv.org/4.x/d2/de8/group__core__array.html

- OpenCV `phaseCorrelate`: https://docs.opencv.org/4.x/d7/df3/group__imgproc__motion.html
- scikit-image metrics (PSNR, SSIM): <https://scikit-image.org/docs/stable/api/skimetrics.html>